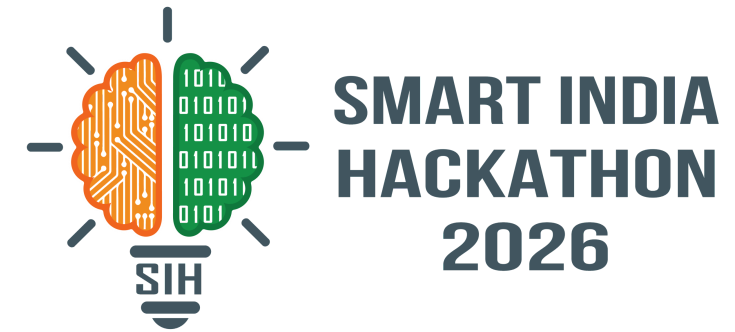


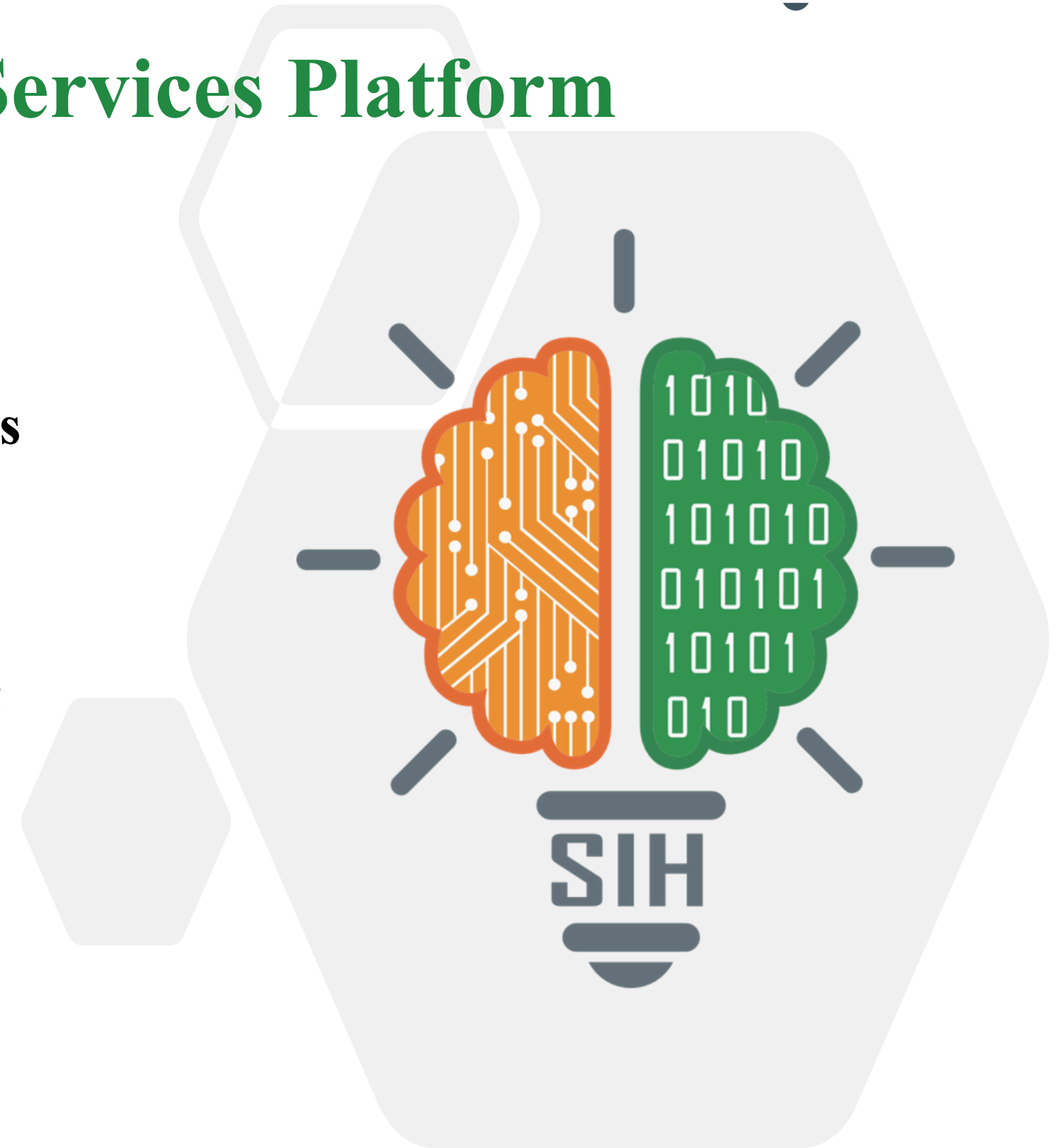


SMART INDIA HACKATHON 2026



AI-Powered Cooperative Gig Services Platform

- **Problem Statement ID** –SIH26089
- **Problem Statement Title**-Cooperative Gig Services
Platform for Household & Community Services
- **Theme**-Agriculture, FoodTech & Rural Development
- **PS Category**- Software
- **Team ID**-
- **Team Name** (Registered on portal)



PROBLEM STATEMENT:

Labour cooperatives have many skilled workers, but there is no proper digital platform to connect them with customers. Private platforms currently dominate this market, so cooperative workers are underutilized. Our solution is to create a cooperative-owned digital marketplace that connects verified workers with households and institutions while ensuring fair wages, worker welfare and consumer trust.

KEY CHALLENGES:

- **Finding Reliable Workers** – Customers often find it difficult to identify skilled and trustworthy workers in their area.
- **Fair Pricing and Wages** – Customers may not know the right service cost, while workers may not always receive fair payment.
- **Better Work Opportunities** – Many cooperative workers have the skills but struggle to reach more customers and find regular work.

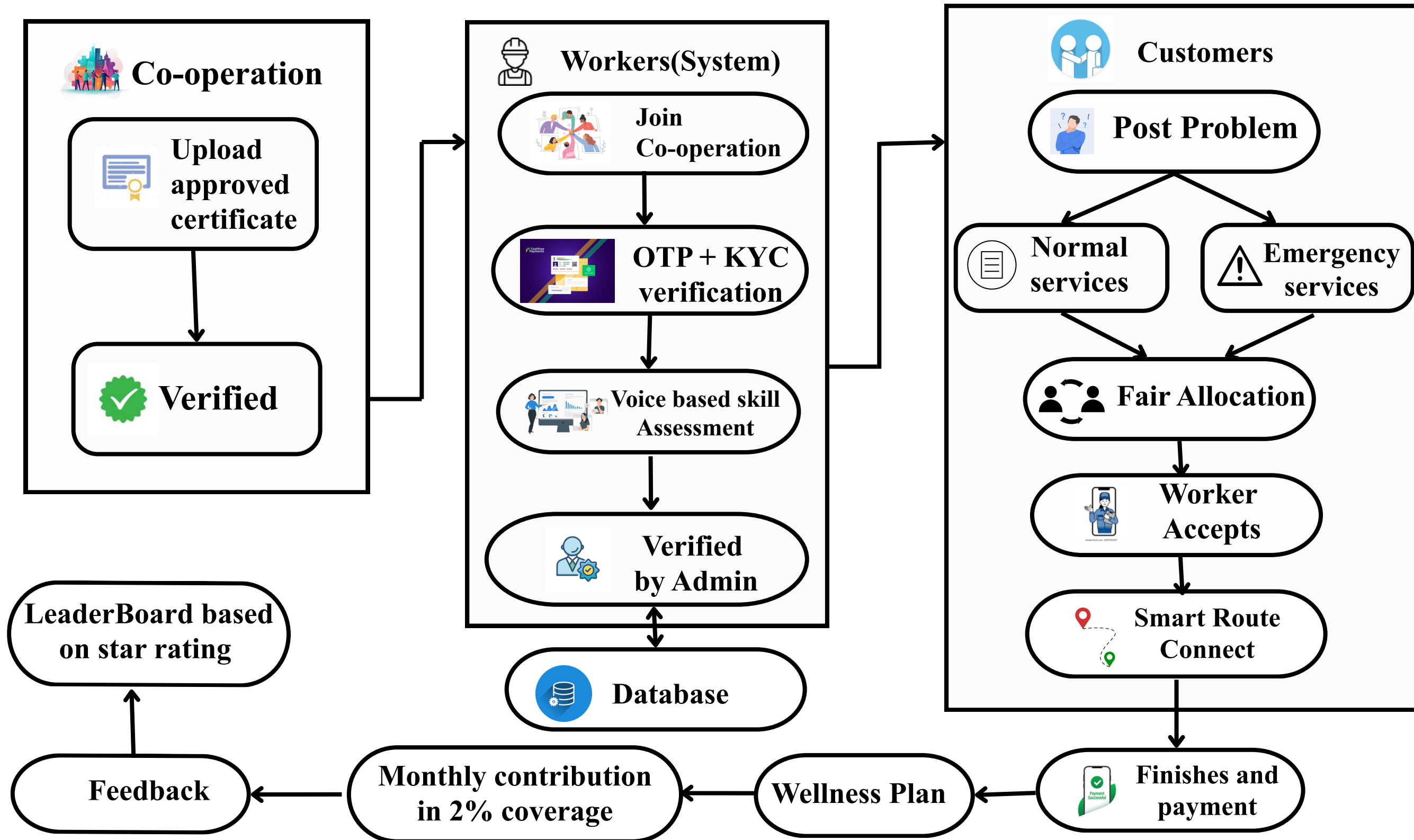
NOVELTY:

- **Fairness-Aware AI Allocation**– Assigns jobs fairly based on skills, performance, and workload.
- **Opportunity Gap Score** – Gives priority to skilled workers with fewer job opportunities.
- **Demand-to-Skill Intelligence** – Predicts demand and identifies skills needed for future jobs.

SOLUTION:

- A cooperative-owned digital service marketplace that connects **verified local workers** with households and institutions.
- Workers can register, create skill profiles, and **get verified** through their Labour **Cooperative Society**.
- Customers can easily search, book, and schedule services based on location, availability, and skills.
- The platform supports **digital payments**, invoices, **ratings**, **feedback**, and **emergency service requests**.
- A cooperative dashboard helps manage workers, bookings, payments, and welfare activities.
- AI can forecast service **demand and support** better workforce allocation.
- **Multilingual** support makes the platform accessible to both workers and customers.

TECHNICAL APPROACH



Technological stack



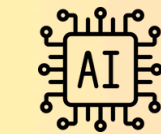
Frontend
Next.js + TypeScript



Backend
FastAPI + Python



Database
PostgreSQL + pgvector



AI / ML
XGBoost + Scikit-learn



Payment Process through
Razor Pay



Location & Routing
Google Maps Platform



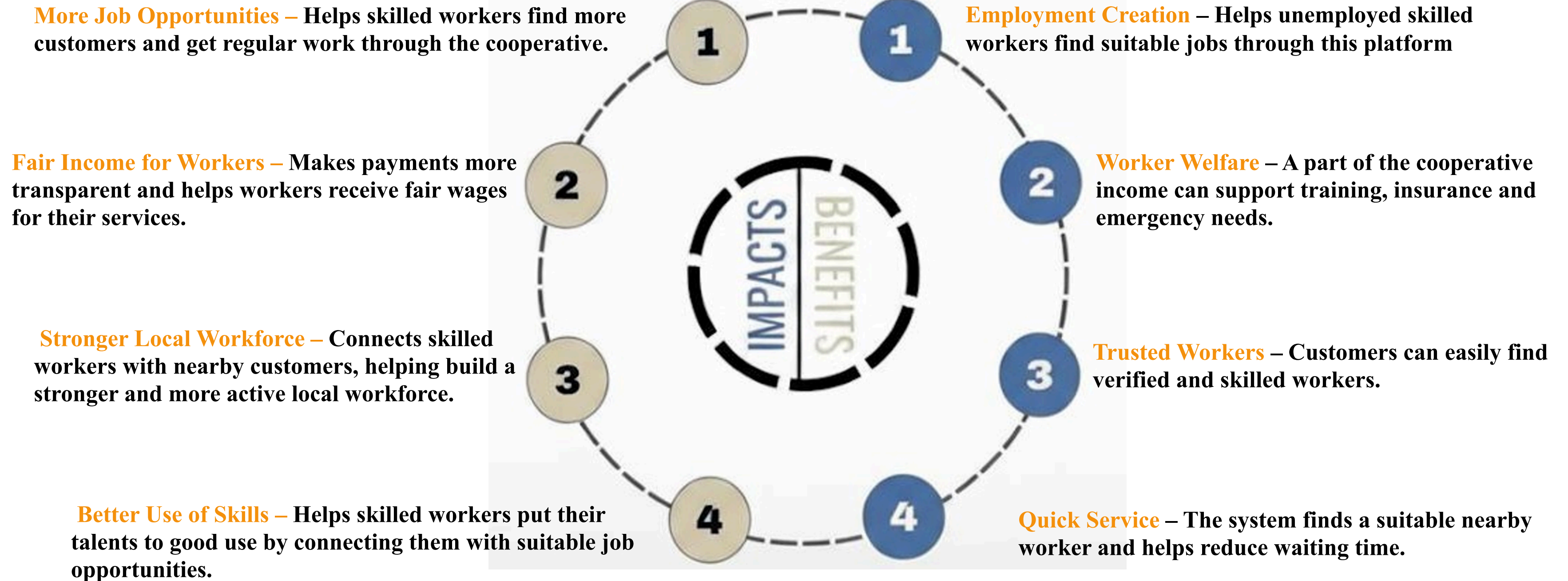
Voice Assessment
Speech-to-Text AI

FEASIBILITY:

- **Technical Feasibility** – The platform can be built using reliable and widely used technologies such as Next.js, FastAPI, PostgreSQL, AI/ML, and Google Maps API.
- **Operational Feasibility** – Cooperatives can easily manage worker verification, service bookings, payments, and welfare activities through one centralized dashboard.
- **Financial Feasibility** – Using open-source technologies and affordable cloud services helps keep development and maintenance costs low.
- **Scalability** – The platform can begin with a single cooperative and gradually expand to more cooperatives, cities, workers, and service categories.

VIABILITY:

- **Sustainable Revenue Model** – A small service fee from each completed booking can help the cooperative maintain and run the platform.
- **Worker Benefits** – Better job allocation and welfare support can help workers get more regular work and improve their income security.
- **Customer Value** – Verified workers, clear pricing, digital payments, and reliable services make customers feel more confident using the platform.
- **Long-Term Growth** – AI-based demand forecasting can help cooperatives understand future service needs and develop the right workforce accordingly.



RESEARCH AND REFERENCES

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- Platform Cooperatives and Employment – OECD – www.oecd.org/en/publications/platform-cooperatives-and-employment_3eab339f-en.html
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